



Serial Port Server (AT Command Set)
(NE2 series)
(Serial port $\rightleftharpoons$) Ethernet



Contents

Chapter 1 "Basic Functions" AT Command Set	1
1.1 Summary of Basic Configuration Commands	2
1.2 Enter AT command	3
1.3 Exit AT Command	3
1.4 Query Model	3
1.5 Query/Set Name	4
1.6 Query/Set ID	4
1.7 Restart	4
1.8 Factory Reset	5
1.9 Query version information	5
1.10 Query/Set Serial Port	5
1.11 Query MAC address	6
1.12 Query/Set Network Parameters	6
1.13 Query/Set Local Port Number	7
1.14 Query/Set Local Operating Mode and Target Device Network Parameters	7
1.15 Query network connection status	8
1.16 Query/Set Serial Port Cache Clearing Status	8
1.17 Query/Set Registration Package Mode	8
1.18 Query/Set Custom Registration Package Contents	9
1.19 Query/Set Network Heartbeat Packet	10
1.20 Query/Set Serial Port Heartbeat Packet	10
1.21 Query/Set Short Connection Time	11
1.22 Query/Set Timeout Restart Time	11
1.23 Query/Set Network Disconnection Reconnection Time	11
1.24 Protocol Distribution	12
1.25 Fast Network AT	12
1.26 AT Configuration Password Enable	13
Chapter 2 "Modbus Functions" AT Instruction Set	14
2.1 Summary of "Modbus Function" Commands	14
2.2 Query Modbus address filtering	14
2.3 Query Modbus operating mode and command timeout	14
2.4 Start Modbus TCP to Modbus RTU protocol conversion	15
2.5 Set the Modbus gateway command storage time and automatic query interval	15
2.6 Querying and Editing Pre-stored Commands for Modbus Configuration Gateways	16
Chapter 3 "Internet of Things Functions" AT Instruction Set	17
3.1 Summary of "Internet of Things Function" Commands	17
3.2 MQTT and HTTP Target IP or Domain Name Configuration	17
3.3 Query/Set HTTP Request Method	17
3.4 Query/Set HTTP URL Path	18
3.5 Query/Set HTTP Header	18
3.6 Query/Set Up MQTT Target Platform	19
3.7 Query/Set MQTT Keep-Alive Heartbeat Packet Sending Period	19

3.8 Query/Set MQTT Device Name (Client ID)	20
3.9 Query/Set MQTT Username (User Name / Device Name)	20
3.10 Query/Set MQTT Product Password (MQTT password/Device Secret)	21
3.11 Querying/Setting Up MQTT Subscription Topics	21
3.12 Query/Set Up MQTT Publish Topic	22
Chapter 4 AT Configuration Examples	26
4.1 Connection Standard MQTT 3.1.1 Server Example	26
Revision History	27
About Us	27

"Basic Functions" AT Command Set

- 1、 To enter AT command mode: Send +++ via serial port, and send AT again within 3 seconds. If the device echoes +OK, then you have entered AT command mode.
- 2、 This instruction manual supports serial port servers of models NE2-D11/NE2-D12/NE2-S1/NE2-T1/NE2-T1B/NE2-T1M, etc.
- 3、 The following text contains “ <CR><LF> ” and “ \r \n ” , which represent newline characters for different text formats. They are actually HEX (0x 0D and 0x 0A).

Error code table:

Error code	illustrate
- 1	Invalid command format
1	Invalid command
2	System error
- 3	Not yet defined
- 4	Invalid parameters
- 5	Incorrect password

1.1 Summary of basic configuration commands

instruction	describe
A T+EXAT	Exit AT configuration mode
A T+MODEL	Equipment Model
AT+NAME	Equipment Name
A T+SN	Device ID
AT+REBT	Restart the device
AT+RESTORE	Factory reset
AT+VER	Check firmware version
AT+UART	Serial port parameters
A T+MAC	Device MAC address
AT+WAN	Device network parameters
AT+LPORT	Device Port
A T+SOCK	Operating mode and target network parameters
AT+LINKSTA	Connection status feedback
AT+UARTCLR	Connect to serial port buffer mode
AT+REGMOD	Registration package mode
AT+REGINFO	Registration package contents
AT+HEARTMOD	Heartbeat Pack Mode
AT+HEARTINFO	Heartbeat Package Contents
A T+SHORTM	short connection
AT+TMORST	Timeout restart
A T+TMOLINK	Disconnect and restart
AT+PASSEN	AT configuration password enabled

1.2 Enter AT command

instruction	AT
Function	Enter AT command mode
send	AT
return	<CR><LF>+OK<CR><LF> / <CR><LF>+OK =AT e nable <CR><LF>
Remark	When there is no connection and no configuration is provided, the following response is returned: + OK=AT enable Returns " +OK" when a connection is established.

[Example]

+++ first, without a newline.

No newline is needed when sending AT .

Received\ r\n+OK\r\n or\ r\n + OK=AT enable \r\n

1.3 Exit AT commands

instruction	AT+ EXAT
Function	Enter AT command mode
send	AT+ EXAT <CR><LF>
return	<CR><LF>+OK<CR><LF>

[Example]

Send: A T+EXAT \ r\n

Received: \ r\n+OK\r\n

1.4 Model Inquiry

instruction	AT+ MODEL
Function	Model Inquiry
send	AT+MODEL<CR><LF>
return	<CR><LF>+OK= < Model String><CR><LF>
Remark	Modelstring:NE2-D11 NE2-D12 NE2-S1 NE2-T1 NE2-T1B

[Example]

Send: A T+MODEL \ r\n

Received: \ r\n +OK= NE2-D11 \r\n

1.5 Query/Set Name

instruction	AT+ NAME
Function	Query and set name
Send (query)	AT+NAME<CR><LF>
Return (query)	<CR><LF>+OK=< N ame String><CR><LF>
Send (Settings)	AT +NAME = < Name String><CR><LF> (Limited to 10 bytes)
Return (Settings)	<CR><LF>+OK<CR><LF>

[Example]

Query:

Send: A T+NAME \r\n

Received: \r\n +OK= admin \r\n

set up:

Send: A T+NAME=001 \r\n

Received: \r\n +OK \r\n

1.6 Query/Set ID

instruction	AT+ SN
Function	Query and set ID
Send (query)	AT+SN<CR><LF>
Return (query)	<CR><LF>+OK=< SN String><CR><LF>
Send (Settings)	AT+SN = < SN String><CR><LF> (Limited to 2-4 bytes)
Return (Settings)	<CR><LF>+OK<CR><LF>

[Example]

Query:

Send: A T+SN \r\n

Received: \r\n +OK=0001\r\n

set up:

Send: A T+SN=111 \r\n

Received: \r\n +OK \r\n

1.7 Restart

instruction	AT+ REBT
Function	Restart
send	AT+REBT<CR><LF>
return	<CR><LF>+OK<CR><LF>

[Example]

Send: A T+ REBT \r\n

Received: \r\n +OK \r\n

Waiting for the restart to complete.

1.8 Factory reset

instruction	AT+ RESTORE
Function	Factory reset
send	AT+RESTORE<CR><LF>
return	<CR><LF>+OK<CR><LF>

[Example]

Send: A T+ RESTORE\r\n

Received: \r\n +OK \r\n

Waiting for the restart to complete.

1.9 Query version information

instruction	AT+ VER
Function	Query version information
send	AT+VER<CR><LF>
return	<CR><LF>+OK<CR><LF>

[Example]

Send: A T+ VER \r\n

Received: \r\n +OK =9167-0-xx \r\n

1.10 Query/Set Serial Port

instruction	AT+ UART
Function	Query and configure serial port
Send (query)	AT+UART<CR><LF>
Return (query)	<CR><LF>+OK=< Baud , Data, Stop, P arity, packetTime,packetLength><CR><LF>
Send (Settings)	AT+UART=< Baud , Data , Stop, P arity, packetTime,packetLength><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	Baud rate : 600, 1200 , 2400, 4800, 9600, 19200, 38400, 57600, 115200, 230400, 460800; Data bits : 8 , 7, 6, 5 Stop (stop bit) : 1 (1 bit), 2 (1.5 bits), 3 (2 bits) Parity : 0 (NONE), 2 (EVEN) , 3 (ODD) packetTime (frame interval) : 1-125 (byte idle time) packetTime (frame length) : 10-1024 bytes

[Example]

Query:
 Send: A T+ UART \ r\n
 Received:\r\n+OK=115200,8,1,0,1,1024\r\n
 set up:
 Send: A T+ UART= 115200,8,1,0,1,1024\ r\n
 Received:\r\n+OK\r\n

1.11 Query MAC address

instruction	AT+ MAC
Function	Query MAC address
send	AT+MAC<CR>
return	<CR><LF>+OK=< MAC ><CR><LF>
Remark	The returned data format is "xx -xx-xx-xx-xx-xx ".

[Example]
 Send: A T+ MAC\ r\n
 Received:\r\n+OK=84 - C2 - E4 - 36 - 05 - A2\r\n

1.12 Query/Set Network Parameters

instruction	AT+ WAN
Function	Query and set network parameters
Send (query)	AT+WAN<CR><LF>
Return (query)	<CR><LF>+OK=< Mode , Address , M ask, Gateway , DNS 1,DNS2><CR><LF>
Send (Settings)	AT+WAN=< Mode , Address , M ask, Gateway , DNS 1,DNS2><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	Mode: DHCP/STATIC Address : Local IP address M ask : Subnet mask Gateway DNS 1 : Primary DNS Server DNS 2 : Alternate DNS server

[Example]
 Query:
 Send: A T+ WAN\ r\n
 Received:\r\n+OK= STATIC ,
 192.168.3.7,255.255.255.0,192.168.3.1,114.114.114.114,8.8.8.8\r\n
 Settings: (Dynamic IP)
 Send: A T+ WAN=DHCP , 192.168.3.7,255.255.255.0,192.168.3.1,114.114.114.114,8.8.8.8\r\n
 Received:\r\n+OK\r\n
 Configuration: (Static IP)

Send: AT + WAN=STATIC , 192.168.3.7,255.255.255.0,192.168.3.1,114.114.114.114,8.8.8.8\r\n
Received:\r\n+OK\r\n

1.13 Query/Set Local Port Number

instruction	AT+ LPORT
Function	Query and set the local port number
Send (query)	AT+LPORT=<linkId><CR>
Return (query)	<CR><LF>+OK=<linkId, Value ><CR><LF>
Send (Settings)	AT+LPORT=<linkId, Value ><CR>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	linkId (link number): 0 or 1 Port number : 0-65535,0 (A random port is used in client mode, and a non-zero parameter is required in server mode, otherwise the device server will fail to start) ;

[Example]

Query:

Send: A T+ LPORT=0\r\n

Received:\r\n+OK=0,8887\r\n

set up:

Send: A T+ LPORT =0, 888 3 \r\n

Received:\r\n+OK\r\n

1.14 Query/set local machine operating mode and target device network parameters

instruction	AT+ SOCK
Function	Query and set network protocol parameters
Send (query)	AT+SOCK=<linkId><CR><LF>
Return (query)	<CR><LF>+OK=<linkId, Model , R e mote IP, R e mote P ort><CR><LF>
Send (Settings)	AT+SOCK=<linkId, Model , R e mote IP, R e mote P ort><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	linkId (link number): 0 or 1 Model (Working Mode) : DISABLE, TCPC, TCPS, UDPC, UDPS, MQTTC, HTTPC ; Remote IP (target IP /domain name): Maximum domain name configuration of 128 characters ; Remote Port (Target Port): 1-65535 ;

[Example]

Query:
 Send: A T+ SOCK=0\ r\n
 Received:\r\n+OK=0,TCPC,192.168.3.3,8888\r\n
 set up:
 Send: AT+SOCK=0,TCPC,192.168.3.100,8886\ r\n
 Received:\r\n+OK\r\n

1.15 Query network connection status

instruction	AT+ LINKSTA
Function	Query network connection status
send	AT+LINKSTA=<linkId><CR><LF>
return	<CR><LF>+OK=<linkId, STA ><CR><LF>
Remark	STA : Connect/Disconnect

[Example]
 Send: A T+ LINKSTA=0\ r\n
 Received:\r\n+OK=0,Disconnect\r\n

1.16 Query/Set Serial Port Cache Clearing Status

instruction	AT+ UARTCLR
Function	Query and set serial port cache clearing status
Send (query)	AT+UARTCLR=<linkId><CR><LF>
Return (query)	<CR><LF>+OK=<linkId, STA ><CR><LF>
Send (Settings)	AT+UARTCLR=<linkId, STA ><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	linkId (link number): 0 or 1 STA : ON (Enable connection, clear cache) OFF (Disable connection, clear cache)

[Example]
 Query:
 Send: A T+ UARTCLR=0\ r\n
 Received:\r\n+OK=0,ON\r\n
 set up:
 Send: A T+ UARTCLR=0,OFF\ r\n
 Received:\r\n+OK\r\n

1.17 Query/Set Registration Package Mode

instruction	AT+ REGMOD
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Function	Query and set registration package mode
Send (query)	AT+REGMOD=<linkId><CR><LF>
Return (query)	<CR><LF>+OK=<linkId, Status ><CR><LF>
Send (Settings)	AT+REGMOD=<linkId, status ><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	linkId (link number): 0 or 1 Status: OFF - Disable OLMAC - Send MAC address upon first connection OLSN - Send SN upon first connection OLCSTM - Send Custom Initial Connection EMBMAC - Send MAC per packet EMBSN - SN sent per packet EMBCSTM - Custom per packet sending OLTIME - First connection sent timestamp EMBTIME - Timestamp of each package sent

[Example]

Query:

Send: A T+ REGMOD=0 \r\n

Received:\r\n+OK=0,OFF\r\n

set up:

Send: A T+ REGMOD=0, OLMAC \r\n

Received:\r\n+OK\r\n

1.18 Query/Set Custom Registration Package Contents

instruction	REGINFO
Function	Query and set custom registration package contents
Send (query)	AT+HEARTINFO=<linkId><CR><LF>
Return (query)	<CR><LF>+OK=<linkId, Mode , Data ><CR><LF>
Send (Settings)	AT+HEARTINFO=<linkId, Mode , Data ><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	linkId (link number): 0 or 1 Module : Data format (HEX) 1 hexadecimal , (STR) string; Data type: ASCII is limited to 128 bytes, HEX is limited to 128 bytes;

[Example]

Query:

Send: A T+ REGINFO=0 \r\n

Received:\r\n+OK=0, STR, regist msg\r\n

set up:

Send: A T+ REGINFO =0, STR, EBYTE TEST \r\n

Received:\r\n+OK\r\n

1.19 Query/Set up network heartbeat packets

instruction	AT+ NETHEARTPKT
Function	Query and set up network heartbeat packets
Send (query)	AT+ NETHEARTPKT=<linkId><CR><LF>
Return (query)	<CR><LF>+OK=<linkId, Mode , Time ,Type, Data ><CR><LF>
Send (Settings)	AT+NETHEARTPKT=<linkId, Mode , Time ,Type, Data ><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	linkId (link number): 0 or 1 Mode: 0 (Off), 1 (SN), 2 (MAC), 3 (Custom data); Time : Time 1 - 65535 s; Type: Data format (HEX) 1 hexadecimal , (STR) string; Data type: ASCII is limited to 128 bytes, HEX is limited to 128 bytes;

[Example]

Query:

Send: A T+ NETHEARTPKT=0 \ r\n

Received:\r\n+OK=0,1,5,STR , net heart packet\r\n

Send: A T+ NETHEARTPKT =0,1,5,STR , net heart packet\ r\n

Received:\r\n+OK\r\n

1.20 Query/Set Serial Port Heartbeat Packet

instruction	AT+ UARTHEARTPKT
Function	Query and set up network heartbeat packets
Send (query)	AT+ UARTHEARTPKT<CR><LF>
Return (query)	<CR><LF>+OK=< Mode , Time ,Type, Data ><CR><LF>
Send (Settings)	AT+UARTHEARTPKT=< Mode , Time , Type, Data ><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	Mode: 0 (Off), 1 (SN), 2 (MAC), 3 (Custom data); Time : Time 1 - 65535 s; Type: Data format (HEX) 1 hexadecimal , (STR) string; Data type: ASCII is limited to 128 bytes, HEX is limited to 128 bytes;

[Example]

Query:

Send: A T+ UARTHEARTPKT=0 \ r\n

Received:\r\n+OK=1,5,STR , net heart packet\r\n

Send: A T+ UARTHEARTPKT =0,1,5,STR , net heart packet\ r\n

Received:\r\n+OK\r\n

1.21 Query/Set Short Connection Time

instruction	AT+ SHORTM
Function	Query and set short connection time
Send (query)	AT+SHORTM=<linkId><CR><LF>
Return (query)	<CR><LF>+OK=<linkId, Time ><CR><LF>
Send (Settings)	AT+SHORTM=<linkId, Time ><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	linkId (link number): 0 or 1 Time : Limits from 0 to 65535, 0 is off;

[Example]

Query:

Send: AT+ SHORTM=0 \r\n

Received:\r\n+OK=0,0\r\n

set up:

Send: A T+ SHORTM =0,5\r\n

Received:\r\n+OK\r\n

1.22 Query/Set Timeout Restart Time

instruction	AT+ TMORST
Function	Query and set timeout restart time
Send (query)	AT+TMORST<CR><LF>
Return (query)	<CR><LF>+OK=< Time ><CR><LF>
Send (Settings)	AT+TMORST=<Time> <CR> <LF> (Limits time to 60-65535s, 0 disables)
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	Time : Limited to 60-65535s, 0 is off;

[Example]

Query:

Send: AT+ TMORST \r\n

Received:\r\n+OK=300\r\n

set up:

Send: A T+ SHORTM = 350\r\n

Received:\r\n+OK\r\n

1.23 Query/Set Network Reconnection Time

instruction	AT+ TMOLINK
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Function	Query and set the network reconnection time.
Send (query)	AT+TMOLINK<CR><LF>
Return (query)	<CR><LF>+OK=< Times ><CR><LF>
Send (Settings)	AT+TMOLINK= <Times> <CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	Times (Network Reconnection Time) : Limited to 10-65535, 0 for fast reconnection;

[Example]

Query:

Send: AT+ TMOLINK \ r\n

Received:\r\n+OK=5\r\n

set up:

Send: A T+ TMOLINK =10\ r\n

Received:\r\n+OK\r\n

1.24 Protocol distribution

instruction	AT+ DISTRIBUTE
Function	Query and configure protocol distribution enable
Send (query)	AT+DISTRIBUTE <CR><LF>
Return (query)	<CR><LF>+OK= <DIST> <CR> <LF>
Send (Settings)	AT+ DISTRIBUTE =< DIST ><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	DIST : 0 - Disable protocol distribution; 1 - Enable protocol distribution

[Example]

Query:

Send: AT+ DISTRIBUTE \ r\n

Received:\r\n+OK=0\r\n

set up:

Send: A T+ DISTRIBUTE =1\ r\n

Received:\r\n+OK\r\n

1.25 Network Fast AT

instruction	AT+NETAT
Function	Query and configure network quick AT
Send (query)	AT+NETAT <CR><LF>
Return (query)	<CR><LF>+OK=<SW , HEAD > <CR><LF>
Send (Settings)	AT+ DISTRIBUTE =<SW , HEAD><CR><LF>

Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	SW : 0 - Disable network fast AT; 1 - Enable network fast AT

[Example]

Query:

Send: AT+ NETAT \ r\n

Received:\r\n+OK=0,NETAT\r\n

set up:

Send: A T+ NETAT =1,NETAT\ r\n

Received:\r\n+OK\r\n

Quick examples of using AT

[Example]

Query:

Send: NETAT+NAME \ r\n

Received:\r\n+OK=admin\r\n

set up:

Send: NETAT+NAME =admin\ r\n

Received:\r\n+OK\r\n

1.26 AT configuration password enabled

instruction	AT+PASSEN
Function	Query and set AT configuration password enable
Send (query)	AT+PASSEN <CR><LF>
Return (query)	<CR><LF>+OK=<SW> <CR> <LF>
Send (Settings)	AT+ PASSEN =<SW><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	SW : OFF - Disable AT configuration password enable ON: Enable AT configuration password

[Example]

Query:

Send: AT+ PASSEN \ r\n

Received:\r\n+OK=OFF\r\n

set up:

Send: A T+ PASSEN = ON\ r\n

Received:\r\n+OK\r\n

" Modbus Function" AT Command Set

1.27 Summary of Modbus Function Commands

instruction	describe
A T+MODWKMOD	Mod bus mode
A T+MODPTCL	Protocol Conversion
AT+MODGTWYTM	Storage gateway instruction storage time and query interval
AT+MODCMDEDIT	Modbus RTU instructions pre-stored

1.28 Query Modbus address filtering

instruction	AT +MODADDFLI
Function	Query and set Modbus filter addresses
Send (query)	AT+MODADDFLI <CR><LF>
Return (query)	<CR><LF>+OK = <FilterAddr> <CR> <LF>
Send (Settings)	AT+ MODADDFLI =<FliterAddr><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	FliterAddr: 0-255, 0: No filtering; 1-255: No filtering of packets at this address.

Query:

Send: AT+ MODADDFLI \ r\n

Received:\r\n+OK= 0 \r\n

set up:

Send: AT+ MODADDFLI = 1\ r\n

Received:\r\n+OK\r\n

1.29 Query Modbus operating mode and command timeout.

instruction	AT+MODWKMOD
Function	Query and set Modbus working mode
Send (query)	AT+ MODWKMOD<CR><LF>
Return (query)	<CR><LF>+OK=<Mode >< Time out> <CR><LF>
Remark	Mode: NONE (Disable MODBUS) SIMPL (Simple Protocol Conversion) MULIT (Multi-host mode) STORE (Storage Gateway) CONFIG (Configurable Gateway)

	AUTOUP (Active Upload Mode) Timeout : 0-65535 ;
--	---

Query:

Send: AT+ MODWKMOD \ r\n

Received:\r\n+OK= SIMPL , 100 \r\n

set up:

Send: AT+ MODWKMOD =MULIT,1000 \ r\n

Received:\r\n+OK\r\n

1.30 Start Modbus TCP to Modbus RTU protocol conversion

instruction	AT+MODPTCL
Function	Query and configure protocol conversion (Modbus) T CP <=> Modbus R TU)
Send (query)	AT+ MODPTCL <CR><LF>
Return (query)	<CR><LF>+OK=<Mode> <CR> <LF>
Remark	Mode: ON (Enable protocol conversion) OFF (Disable protocol conversion)

Query:

Send: AT+ MODPTCL \ r\n

Received:\r\n+OK= ON \r\n

set up:

Send: AT+ MODPTCL = ON \ r\n

Received:\r\n+OK\r\n

1.31 Configure the Modbus gateway command storage time and automatic query interval.

instruction	AT+MODGTWYTM
Function	Query and configure the storage time of Modbus gateway commands and the automatic query interval.
Send (query)	AT+ MODGTWY TM <CR><LF>
Return (query)	<CR><LF>+OK=< Time 1 > < Time2> <CR><LF>
Remark	Time 1 : Instruction storage time (1-255 seconds) Time2: Automatic query interval (1 - 65535 milliseconds)

Query:

Send: AT+ MODGTWYTM \ r\n

Received:\r\n+OK= 10 , 200 \r\n

set up:

Send: AT+ MODGTWYTM = 5 , 100 \ r\n

Received:\r\n+OK\r\n

1.32 a Modbus configuration gateway

instruction	AT+MODCMDEDIT
Function	a Modbus configuration gateway
Send (query)	AT+ MODCMDEDIT <CR><LF>
Return (query)	<CR><LF>+OK= <Mode> <CMD> <CR> <LF>
Remark	Mode : Add command ; DEL delete command ; CLR clear command; C MD : Modbus commands (only standard Modbus RTU commands are supported, no checksum is required, and only function codes 01, 02, 03, and 04 can be configured to be read) . The same command cannot be stored; otherwise, return +ERR =-4 .

Query:

Send: AT+ MODCMDEDIT \ r\n

Received:\r\n+OK=\r\n

1: 02 03 00 00 00 02\r\n

2: 01 03 00 05 00 00\r\n

set up:

Send: AT+ MODCMDEDIT = ADD,0103000A0003 \ r\n (Add command)

Received:\r\n+OK\r\n

Send: AT+ MODCMDEDIT = DEL,0103000A0003 \ r\n (Delete command)

Received:\r\n+OK\r\n

Send: AT+ MODCMDEDIT = CLR,0103000A0003 \ r\n (Clear command)

Received:\r\n+OK\r\n

"Internet of Things (IoT) Functionality" AT Instruction Set

1.33 Summary of "Internet of Things Function" Commands

instruction	describe
AT+HTPREQMODE	HTTP request method
AT+HTPURL	HTTP URL path
AT+HTPHEAD	HTTP headgear
AT+MQTTCLOUD	MQTT platform
AT+MQTKPALIVE	MQTT heartbeat survival cycle
AT+MQTDEVID	MQTT Client ID
AT+MQTUSER	MQTT User Name
AT+MQTPASS	MQTT Password
AT+MQTTPRDKEY	Alibaba Cloud Product Key
AT+MQTSUB	MQTT Subscription Topic
AT+MQTPUB	MQTT release theme

1.34 MQTT and HTTP target IP or domain name configuration

Refer to "Query/Set Local Operating Mode and Target Device Network Parameters".

1.35 Query/Set HTTP Request Method

instruction	AT+HTPREQMODE
Function	Query and configure HTTP client mode request methods
Send (query)	AT+HTPREQMODE=<linkId><CR><LF>
Return (query)	<CR><LF>+OK=<linkId, Method><CR><LF>
Send (Settings)	AT+HTPREQMODE=<linkId, Method><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	linkId (link number): 0 or 1 Method : GET \ POST

[Example]

Query:

Send: AT+HTPREQMODE \ r\n

Received: \r\n+OK=0, GET \r\n

set up:

Send: AT+HTPREQMODE=0, POST \ r\n

Received: \r\n+OK\r\n

1.36 Query/Set HTTP URL Path

instruction	AT+HTPURL
Function	Query and set HTTP URL paths
Send (query)	AT+ HTPURL =<linkId><CR><LF>
Return (query)	<CR><LF>+OK=<linkId, P ath><CR><LF>
Send (Settings)	AT+ HTPURL =<linkId, Path ><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	Path: HTTP request URL resource address (length limited to 0-256 characters)

[Example]

Query:

Send: AT+ HTPURL =0 \r\n

Received: \r\n+OK=0, /1.php? \r\n

set up:

Send: A T+ HTPURL =0, /view/ed7e65a90408763231126edb6f1aff00bfd57061.html \r\n

Received: \r\n+OK\r\n

1.37 Query/Set HTTP Headers

instruction	AT+HTPHEAD
Function	Query and set HTTP headers
Send (query)	AT+ HTPHEAD =<linkId><CR><LF>
Return (query)	<CR><LF>+OK= < linkId, Para , Head ><CR><LF>
Send (Settings)	AT+ HTPHEAD = < linkId, Para , Head ><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	linkId (link number): 0 or 1 Para(Whether the HTTP return serial port data includes a header) : D EL: Without a cover; A DD: with a cap; Header (HTTP request header) : Length limit 256 characters;

[Example]

Query:

Send: AT+ HTPHEAD =0 \r\n

Received: \r\n+OK=0, DEL,User-Agent: Mozilla/5.0 \r\n

set up:

Send: A T+ HTPHEAD =0, ADD, Host:www.ebyte.com \r\n

Received: \r\n+OK\r\n

1.38 Query/Set up MQTT target platform

instruction	AT+MQTTCLOUD
Function	Query and configure the MQTT target platform
Send (query)	AT+MQTTCLOUD=<linkId><CR><LF>
Return (query)	<CR><LF>+OK=<linkId, Server><CR><LF>
Send (Settings)	AT+MQTTCLOUD=<linkId, Server><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	linkId (link number): 0 or 1 Server (MQTT target platform) : S TANDARD (MQTT 3.1.1 standard protocol server) OneNET (OneNET -MQTT Server) A LI (Alibaba Cloud MQTT Server) B AIDU (Baidu Cloud M QTT Server) H UAWEI (Huawei Cloud M QTT Server)

[Example]

Query:

Send: AT+MQTTCLOUD=0 \r\n

Received:\r\n+OK=0, STANDARD \r\n

set up:

Send: AT+MQTTCLOUD=0, BAIDU \r\n

Received:\r\n+OK\r\n

1.39 Query/Set MQTT Keep-alive Heartbeat Sending Period

instruction	AT+MQTKPALIVE
Function	Query and set the MQTT keep-alive heartbeat time period
Send (query)	AT+MQTKPALIVE=<linkId><CR><LF>
Return (query)	<CR><LF>+OK=<linkId, Time><CR><LF>
Send (Settings)	AT+MQTKPALIVE=<linkId, Time><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	linkId (link number): 0 or 1 Time: MQTT keep-alive heartbeat time (limited to 30-1200 seconds, default 60 seconds , modification is not recommended);

[Example]

Query:

Send: AT+MQTKPALIVE=0 \r\n

Received:\r\n+OK=0, 60 \r\n

set up:

Send: AT+MQTKPALIVE=0, 30 \r\n

Received:\r\n+OK\r\n

1.40 Query/set MQTT device name (Client ID)

instruction	AT+MQTDEVID
Function	Query and set the MQTT device name (Client ID).
Send (query)	AT+ MQTDEVID =<linkId><CR><LF>
Return (query)	<CR><LF>+OK=<linkId, Client ID ><CR><LF>
Send (Settings)	AT+ MQTDEVID =<linkId, Client ID ><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	linkId (link number): 0 or 1 Client ID : MQTT device name (Client ID) with a length limit of 256 characters;

[Example]

Query:

Send: AT+ MQTDEVID =0 \ r\n

Received:\r\n+OK=0, test-1 \r\n

set up:

Send: A T+ MQTDEVID =0,+OK=0,test-iot\ r\n

Received:\r\n+OK\r\n

1.41 Query/set MQTT username (User Name / Device Name)

instruction	AT+MQTUSER
Function	Query and set MQTT username (User Name/ Device Name)
Send (query)	AT+ MQTUSER =<linkId><CR><LF>
Return (query)	<CR><LF>+OK=<linkId, User Name ><CR><LF>
Send (Settings)	AT+ MQTUSER =<linkId, User Name ><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	linkId (link number): 0 or 1 User Name : MQTT Product ID (User Name / The device name is limited to a length of 256 characters.

[Example]

Query:

Send: AT+ MQTUSER =0 \ r\n

Received: \r\n+OK=0, ebyte-IOT \r\n

set up:

Send: A T+ MQTUSER =0, 12345678&a1Ofdo5l0 \ r\n

Received:\r\n+OK\r\n

1.42 Query/set MQTT product password (MQTT password/Device Secret)

instruction	AT+ M QTPASS
Function	Query and set the MQTT login password (MQTT Password / Device Secret)
Send (query)	AT+ MQTPASS =<linkId><CR><LF>
Return (query)	<CR><LF>+OK=<linkId, Password ><CR><LF>
Send (Settings)	AT+ MQTPASS =<linkId, Password ><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	linkId (link number): 0 or 1 Password: The MQTT login password (MQTT Password/Device Secret) is limited to 256 characters in length.

[Example]

Query:

Send: AT+ MQTPASS =0 \ r\n

Received:\r\n+OK=0, 12345678 \r\n

set up:

Send: A T+ MQTPASS =0, 87654321 \ r\n

Received:\r\n+OK\r\n

1.43 Query/Set up MQTT subscription topics

instruction	AT+MQTSUB
Function	Query and configure MQTT subscription topics
Send (query)	AT+ MQTSUB =<linkId><CR><LF>
Return (query)	<CR><LF>+OK= < linkId, Q o s>, < T opic><CR><LF>
Send (Settings)	AT+ MQTSUB = < linkId, Q o s,Topic ><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	linkId (link number): 0 or 1 QoS : Only supports QoS levels 0 and 1; Topic: MQTT subscription topic (length limit 256 characters)

[Example]

Query:

Send: AT+ MQTSUB =0 \ r\n

Received:\r\n+OK=0, 0,topic \r\n

set up:

Send: A T+ MQTSUB =0, 0,/ggip6zWo8of/TEST/user/SUB \ r\n

Received:\r\n+OK\r\n

1.44 Query/Set up MQTT publish topics

instruction	AT+MQTPUB
Function	Query and configure MQTT publishing topics
Send (query)	AT+ MQTPUB =<linkId><CR><LF>
Return (query)	<CR><LF>+OK= <linkId, Qos,T opic,Cycle><CR><LF>
Send (Settings)	AT+ MQTPUB = <linkId, Qos,Topic ,Cycle><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	linkId (link number): 0 or 1 QoS : Only supports QoS levels 0 and 1; Topic: MQTT publishing topic (length limit 256 characters) Cycle: Release Level 1 retransmission interval (1-10s) (Retransmissions within this interval will occur within 30s)

[Example]

Query:

Send: AT+ MQTPUB =0 \ r\n

Received:\r\n+OK=0, 0,topic \r\n

set up:

Send: A T+ MQTPUB = 0, 0,/ggip6zWo8of/TEST/user/PUB \ r\n

Received:\r\n+OK\r\n

1.45 Query/Set SNTP Function Switch

instruction	AT+SNTPENABLE
Function	Query and set the SNTP function switch status
Send (query)	AT+ SNTPENABLE <CR><LF>
Return (query)	<CR><LF>+OK= <STATE> <CR><LF>
Send (Settings)	AT+ SNTPENABLE = < STATE ><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	STATE: Represents the function's enabled status, 0 or 1.

[Example]

Query:

Send : AT + SNTPENABLE

Received:\r\n+OK=0\r\n

set up:

Send: AT+ SNTPENABLE =1 \ r\n

Received:\r\n+OK\r\n

1.46 Query/Set SNTP Server URL Address

instruction	AT+ SNTPSERVERURL
Function	Query and set the SNTP server URL address
Send (query)	AT+ SNTPSERVERURL <CR><LF>
Return (query)	<CR><LF>+OK= <URL> <CR><LF>
Send (Settings)	AT+ SNTPSERVERURL = < URL ><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	URL: Represents the URL address of the SNTP remote server (length limited to 256 characters).

[Example]

Query:

Send: AT+ SNTPSERVERURL \r\n

Received:\r\n +OK=ntp.ntsc.ac.cn \r\n

set up:

Send: AT+ SNTPENABLE = ntp1.aliyun.com \r\n

Received:\r\n+OK\r\n

1.47 Query/Set SNTP Time Zone Location

instruction	AT+ SNTPTIMEZONE
Function	This command allows you to query and set the system's time zone. Setting this parameter will modify the system time; the device's factory default is UTC+8.
Send (query)	AT+ SNTPTIMEZONE <CR><LF>
Return (query)	<CR><LF>+OK= < TZ_STR><CR><LF>
Send (Settings)	AT+ SNTPTIMEZONE = < TZ_INT ><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	TZ_STR: A UTC format string, ranging from "UTC-12 to UTC+12". TZ_INT: -12 to +12 represents the time zone to be set.

[Example]

Query:

Send: AT+ SNTPTIMEZONE \r\n

Received:\r\n +OK=UTC+8 \r\n

set up:

Send: AT+ SNTPTIMEZONE =1 \r\n

Received:\r\n+OK\r\n

1.48 Query/Set Client Overflow Handling in TCP Server Mode

instruction	AT+ TCPSCONMANGER
Function	Query and configure how to handle situations where the number of client connections exceeds the limit under a TCP server for a specified link.
Send (query)	AT+ TCPSCONMANGER =<LINK> <CR><LF>
Return (query)	<CR><LF>+OK= < LINK ,STATE><CR><LF>
Send (Settings)	AT+ TCPSCONMANGER = < LINK ,STATE><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	LINK: The link number to query/set STATE: Management mode, 0: Deny new client access. 1: The client whose connection was disconnected earliest

[Example]

Query:

Send: AT+ TCPSCONMANGER =1 \r\n

Received:\r\n +OK=1,0 \r\n

set up:

Send: AT+ TCPSCONMANGER =1,1 \r\n

Received:\r\n+OK\r\n

1.49 Query/Set Link TLS Connection Mode

instruction	AT+ TLSMODE
Function	Query and set the TLS function mode switch for a specified link in TCP client/MQTT/HTTP mode.
Send (query)	AT+ TLSMODE =<LINK> <CR><LF>
Return (query)	<CR><LF>+OK= < LINK ,STATE><CR><LF>
Send (Settings)	AT+ TLSMODE = < LINK ,STATE><CR><LF>
Return (Settings)	<CR><LF>+OK<CR><LF>
Remark	LINK: The link number to query/set STATE: TLS connection mode, parameter range and explanation are as follows: 0: Disable TLS 1: Enable TLS but do not verify server certificates. 2: Use a custom root certificate to verify the server certificate. 3: Use a custom root certificate and a custom device certificate for two-way authentication. 4: Use the root certificate that comes with the device to verify the server certificate. 5: Use the device's built-in root certificate and a custom device certificate for two-way authentication.

[Example]

Query:

Send: AT+ TLSMODE =1 \ r\n

Received:\r\n +OK=1,0 \r\n

set up:

Send: AT+ TLSMODE =1,1 \ r\n

Received:\r\n+OK\r\n

Of AT configuration

1.50 of connecting to a standard MQTT 3.1.1 server

```
{
    Client ID: 876275396
    MQTT username: 485233
    MQTT password: E_DEV01
    mqtt server: mqtt.heclouds.com
    mqtt port:6002
}
```

Before configuring, restore factory settings to avoid enabling unused functions.

SEND (+++)

SEND (AT) within 3 seconds

RCV(+OK=AT enable)

SEND (AT+RESTORE)

RCV(+OK)

The above steps can be used to restore factory settings using hardware.

Step 1: Enter AT configuration mode;

SEND (+++)

SEND (AT) within 3 seconds

RCV(+OK=AT enable)

Step 2: Enable dynamic IP . If it is a local area network MQTT server , you can configure the corresponding IP . Here, we use dynamic IP .

SEND(AT+WAN=DHCP, 192.168.3.7,255.255.255.0,192.168.3.1,114.114.114.114,8.8.8.8)

RCV(+OK)

Step 3: Configure the working mode and MQTT server address and port;

SEND(AT+SOCK= 0, MQTTC,mqtt.heclouds.com,6002)

RCV(+OK=And local port has been set to 0)

Step 4: Select the MQTT platform ;

SEND(AT+MQTTCLOUD= 0, STANDARD)

RCV(+OK)

Step 5: Configure the device's Client ID;

SEND(AT+MQTDEVID= 0, 876275396)

RCV(+OK)

Step 6: Configure the device's MQTT username;

SEND(AT+MQTUSER= 0, 485233)

RCV(+OK)

Step 7: Configure the device's MQTT password;

SEND(AT+MQTPASS= 0, E_DEV01)

RCV(+OK)

Step 8: Subscribe to the corresponding topic;

SEND(AT+MQTSUB=0, 0, EBYTE_TEST)

RCV(+OK)

Step 9: Configure the theme to be used for publishing;

SEND(AT+MQTPUB=0, 0, EBYTE_TEST)

RCV(+OK)

Step 10: Restart the device;

SEND(AT+REBT)

RCV(+OK)

Revision History

Version	Revision Date	Revision Notes
1.0	2024-02-28	Initial version
1.1	2024-10-18	New directives related to SNTP and TLS have been added.
1.2	2026-02-27	Fix the command error in section 4.1

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